

# 西南财经大学本科期末考试试题册（A）

（2021—2022学年第2学期）

以下各项由命题教师填写：

课程名称：面向对象程序设计（JAVA EE）（英）命题教师：陈中普

适用对象（年级专业）：2021级信息管理与信息系统

使用试题的任课教师姓名：陈中普

试题说明：

- 1、考试类型：闭卷[ ] 开卷[☒]
- 2、本套试题共 三 道大题，共 4 页，完卷时间 120 分钟。
- 3、考试用品中除纸、笔、尺子外，可另带的用具有：  
计算器[ ] 字典[ ] 等  
(请在下划线上填上具体数字或内容，所选[ ]内打钩)

考试时间（由制卷方填写）：

以下各项由学生填写：

任课教师：

年级专业：

学生姓名：

学 号：

- 考生注意事项：
- 1.出示学生证（或身份证）和准考证于桌面左上角，以备查验。
  - 2.拿到试卷后清点并检查试卷页数，如有重页、页数不足、空白页及印刷模糊等举手向监考教师示意调换试卷。
  - 3.答题前先将试题册及答题纸上的任课教师、专业、年级、学号、姓名填写完整。
  - 4.所有答案均需填写在答题纸上，答在试题册上无效。
  - 5.考生不得携带任何通讯工具进入考场。
  - 6.考试结束后，将试题册、答题纸和草稿纸等下发材料上交监考教师，不得带离考场。
  - 7.严格遵守考场纪律。

**Part One: Multi-choices (2 marks each; 20 marks in total)**

*(Note: Each question has only one correct choice.)*

1. Which of the following statement is correct in terms of Java EE?
  - A. All Java SE code is valid in Java EE.
  - B. Java EE can only be used for Java Web.
  - C. The version of Java EE is consistent with that of Java SE.
  - D. The package name of Java EE starts with *javaee*.
2. Which of the following statement is correct in terms of HTML?
  - A. One can use pure HTML to display current date in the web page.
  - B. `<image>` element is used for display an image.
  - C. `<a src="abc">` is a hyperlink to a source named "abc".
  - D. HTML elements can be nested.
3. Which of the following statement is correct in terms of Servlet?
  - A. It is mainly used to handle HTTP requests and then make a response.
  - B. It usually outputs a JSP as the response.
  - C. One servlet can only be mapped to one URL.
  - D. A servlet can only be used for GET request, or POST request, but not both.
4. Which of the following statement is correct in terms of HTTP?
  - A. An URL always indicates the request method.
  - B. The parameter of POST can be found in the URL.
  - C. GET is used for retrieving resources by design.
  - D. We can use either GET or POST to upload files.
5. How would servlet code from a service method (e.g., *doPost()*) retrieve the value of the "User-Agent" header from the request?
  - A. `String agent = request.getParameter("User-Agent");`
  - B. `String agent = request.getHeader("User-Agent");`
  - C. `String agent = request.getAttribute("User-Agent");`
  - D. `String agent = getServletContext().getInitParameter("User-Agent");`
6. Which is most logically stored as an attribute in session scope?
  - A. A copy of a query parameter entered by a user.
  - B. The result of a database query to be returned immediately to a user.
  - C. An object representing a user who has just logged into the system.
  - D. A copy of an initialization parameter retrieved from a *ServletContext* object.
7. Which of the following statement is correct in terms of init paramters and attributes?
  - A. One can set the request attribute in DD (*web.xml*) file.
  - B. One can set the request attribute in annotations.
  - C. One can use `setInitParameter("name", "value")` to set the init parameters.
  - D. One can only set the string values for init parameters.

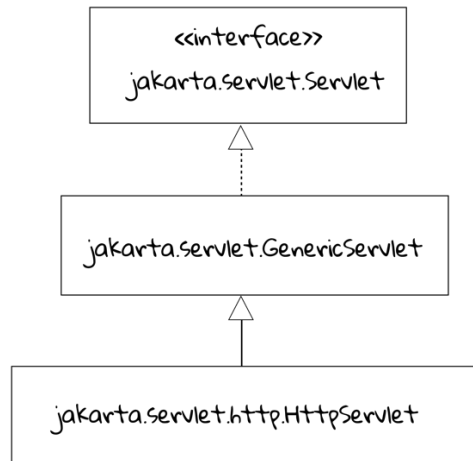
8. Which of the following statement is correct in terms of Session?
- A. A session will always live unless we kill it.
  - B. A session cannot be shared across the browsers.
  - C. Both cookies and sessions are stored in the client.
  - D. If cookies are disabled, then sessions will not work.
9. Suppose there is an object *dog* which has a field “age”, and we put it into the attribute in a servlet: `request.setAttribute(“dog”, dog)`. Then we forward this request to *me.jsp*.
- A. We cannot preview the content of *me.jsp* in a browser directly.
  - B. We must use “dog” as the name of this attribute.
  - C. In *me.jsp*, we can always use `#{dog.age}` to get the age of this dog.
  - D. The attribute still exists even if the user visits *me.jsp* directly.
10. Which of the following statement is correct in terms of Java Web?
- A. JSP is only the choice for View.
  - B. JavaScript is the subset of Java in Web.
  - C. We can store the value of “how many people are online” in a session.
  - D. If a value is stored in a session or context, we shall consider the thread safety.

## Part Two: Short-answer Questions (10 marks + 15 marks + 20 marks)

1. Bob wants to design a web page which enable users to input their ages, and then the server can recommend books given an age. Suppose the code snippet from servlet is:

```
1. @WebServlet(url="/books")
2. public class BookServlet extends HttpServlet {
3.     public void doGet(HttpServletRequest request, HttpServletResponse response) {
4.         int howOld = Integer.parseInt(request.getParameter("age"));
5.         ...
6.     }
7. }
```

Please complete the key HTML code of `<form>` element.



2. Based on the figure above,
  - a) What can you tell from it?
  - b) If someone claims that “GET is used to retrieve a resource, and server would raise an error if we try to update something in *doGet()* method”, how do you think of it?
3. Suppose a list of books are stored in a request’s attribute, and the following is the code snippet:

```

1. List<Book> books = db.getBooksByAge(age);
2. request.setAttribute("books", books);
3. RequestDispatcher dispatcher = request.getRequestDispatcher("books.jsp");
4. dispatcher.forward(request, response);
  
```

*Book* here is a regular Java bean, which contains attributes *name* and *price*.

- a) Please write down the key code to display the prices of returned books using JSTL.
- b) Can we use *sendRedirect()* method instead of *forward()*? Please explain the reason.
- c) Bob thinks it would be better to put *book.jsp* inside *WEB-INF*. Try to defend his idea.
- d) Suppose we find the following CSS code in *home.jsp*. What does *p.price* mean?

```

1. p.price {
2.   color: red;
3. }
  
```

### Part Three: Design (10 marks + 25 marks)

1. Suppose the entry point for College English Test (CET) score query is <https://cjcx.neea.edu.cn/20552>, where 20552 is the number of CET under National Education Examinations Authority (NEEA). One day the IT department of CET decided to develop a new website by adding many new features. What’s more, a fancy and easy-to-remember URL <https://cet.edu.cn/> is used, which renders the previous one obsolete. However, many people have already added the previous one in their bookmarks, so it is likely for them to visit an old and obsolete website.

Please design a solution to this problem.

2. Suppose your teammate has completed the logic for authenticating username/password, and she offers you a public API in *User* class which has a non-arg constructor:

```
1. /// Return true if the name and password are matched;  
2. /// Otherwise return false.  
3. public boolean isValid(String name, String password)
```

You shall use it to authenticate the user. Please complete the code in a servlet based on the process flow. Note that the name can be in non-Latin characters.

(Hint: it is fine to write multiple lines of code for each circle number.)

Start

```
1. ⑥  
2. String user = request.getParameter("name");  
3. String password = request.getParameter("password");  
4. // isValid  
5. if (①) {  
6.     // add to session  
7.     ②  
8.     // go to /center  
9.     ③  
10. } else {  
11.     ④  
12. }
```